Monte Carlo

Informática Gráfica



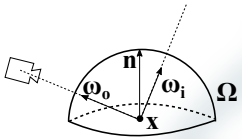
Adolfo Muñoz – Julio Marco

Grado en Ingeniería Informática

Graphics and Imaging Lab

The Render Equation

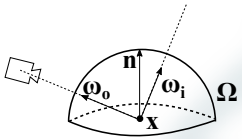
The Render Equation



How do we calculate this?

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i}$$

The Render Equation

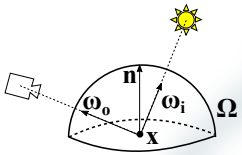


How do we calculate this?

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At least we know how to calculate direct light (delta function).

Direct light

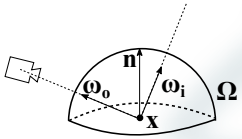


Direct light

Point light in position $\mathbf{c}$ with power p (inverse square law):

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \frac{p}{|\mathbf{c}-\mathbf{x}|^2} f_r(\mathbf{x}, \frac{\mathbf{c}-\mathbf{x}}{|\mathbf{c}-\mathbf{x}|}, \omega_\mathbf{o}) |\mathbf{n} \cdot \frac{\mathbf{c}-\mathbf{x}}{|\mathbf{c}-\mathbf{x}|}|$$

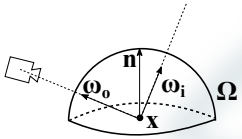
Indirect light



How do we calculate this?

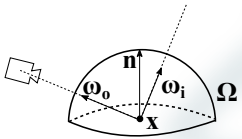
$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i}$$

Indirect light



$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) \quad \underbrace{f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o})|\mathbf{n} \cdot \omega_\mathbf{i}|}_{\text{MATHEMATICAL EXPRESSION}} \quad d\omega_\mathbf{i}$$

Indirect light



$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) \underbrace{f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o})|\mathbf{n} \cdot \omega_\mathbf{i}|}_{\text{MATHEMATICAL EXPRESSION}} d\omega_\mathbf{i}$$

We must **sample** directions ω_i and find L_i from those directions (recursivelly)

Indirect light

Easier with the parametrized version:

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i||\sin\theta_i| d\theta_i d\phi_i$$

Indirect light

Easier with the parametrized version:

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

1. Sample θ_i and ϕ_i.

Indirect light

Easier with the parametrized version:

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

1. Sample θ_i and ϕ_i.
2. Calculate $\omega_\mathbf{i}$ and therefore $f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o})$.

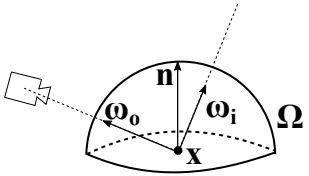
Indirect light

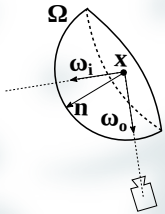
Easier with the parametrized version:

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

1. Sample θ_i and ϕ_i.
2. Calculate $\omega_\mathbf{i}$ and therefore $f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o})$.
3. Find $L_i(\mathbf{x}, \omega_\mathbf{i})$: cast a new ray and get $L_o(\mathbf{x}, \omega_\mathbf{o})$ recursively

Local coordinates





- Local coordinates to global coordinates through an orthonormal change of base $\mathbf{T}$

Local coordinates

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

We need $\omega_\mathbf{i}$. We first define a **local coordinate system**:

- One axis being the normal ($\mathbf{z} = \mathbf{n}$).
- The other two axis ($\mathbf{x}$ and $\mathbf{y}$) being tangent (perpendicular to the normal). Any two of them suffice (unless tangents are defined for the model or useful).

Local coordinates

Local to global transform matrix $\mathbf{T}$:

$$\mathbf{T} = \begin{pmatrix} u_x & v_x & n_x & x_x \\ u_y & v_y & n_y & x_y \\ u_z & v_z & n_z & x_z \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

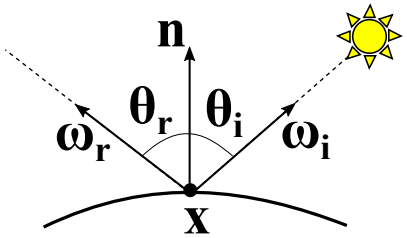
Calculate local direction $\omega_{\mathbf{i}}'$ and change to global coordinates:

$$\omega_{\mathbf{i}}' = \begin{pmatrix} \sin\theta_i \cos\phi_i \\ \sin\theta_i \sin\phi_i \\ \cos\theta_i \end{pmatrix} \qquad \omega_{\mathbf{i}} = \mathbf{T}\omega_{\mathbf{i}}'$$

Local coordinates

Idea: Some maths are easier in local coordinate space.

Law of reflection



$$\omega'_{\mathbf{r}} = \begin{pmatrix} -\omega'_{ix} \\ -\omega'_{iy} \\ \omega'_{iz} \end{pmatrix}$$

$$\omega_{\mathbf{r}} = \mathbf{T}\omega'_{\mathbf{r}}$$

Quadrature

Integration

Exercise

Calculate the folowing integral:

$$\int_0^2 2dx$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^2 x dx$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^2 \int_{-1}^1 x dy dx$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^2 \int_{-1}^1 xydydx$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} d\theta d\phi$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^{\pi} \cos\theta d\theta$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^{\pi} \sin\theta d\theta$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^{\pi} \frac{\sin\theta}{\log\theta} d\theta$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

Integration

Exercise

Calculate the folowing integral:

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

This **cannot be calculated analytically**.

Integration

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

This cannot be calculated analytically:

- $f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i|$ is not integrable analytically.
- $L_i(\mathbf{x}, \omega_\mathbf{i})$ is not even an analytic function (new ray).

Integration

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

This cannot be calculated analytically. What can we do?

Integration

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

This cannot be calculated analytically. What can we do?

- We are screwed.

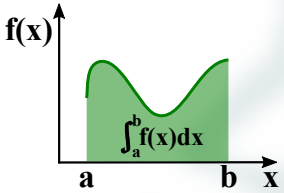
Integration

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

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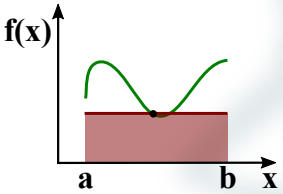
- We are screwed.
- We calculate it numerically (quadrature methods).

Quadrature



- The integral of the function is the area below that function.
- **Idea:** approximate the function so we can approximate its area.

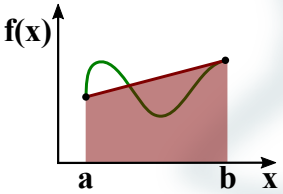
Quadrature



Rectangle rule

$$\int_a^b f(x)dx \approx (b-a)f\left(\frac{a+b}{2}\right)$$

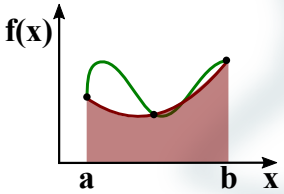
Quadrature



Trapezoidal rule

$$\int_a^b f(x)dx \approx (b-a)\left(\frac{f(a)+f(b)}{2}\right)$$

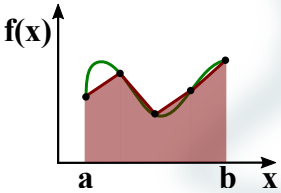
Quadrature



Simpson's rule

$$\int_a^b f(x)dx \approx \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right)$$

Quadrature

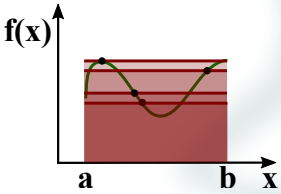


Steps

$$\int_a^b f(x)dx = \int_a^{\frac{3a+b}{4}} f(x)dx + \int_{\frac{3a+b}{4}}^{\frac{a+b}{2}} f(x)dx + \int_{\frac{a+b}{2}}^{\frac{a+3b}{4}} f(x)dx + \int_{\frac{a+3b}{4}}^b f(x)dx$$

$$\int_a^b f(x)dx \approx \frac{(b-a)}{4}\left(f(a) + 2f(3a+b) + 2f(2a+2b) + 2f(a+3b) + f(b)\right)$$

Quadrature

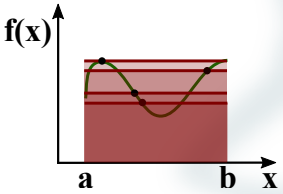


Monte Carlo

$$\int_a^b f(x)dx \approx \frac{b-a}{N}\sum_{i=1}^{N} f(x_i)$$

$N \rightarrow$ number of samples | $x_i \in [a,b] \rightarrow$ uniform random samples

Quadrature



Monte Carlo

$$\int_a^b f(x)dx \approx \frac{b-a}{N}\sum_{i=1}^{N} f(x_i)$$

$N \rightarrow$ number of samples

Yes, **random samples**

$x_i \in [a,b] \rightarrow$ uniform random samples

Quadrature

1 dimension

$$\int_a^b f(x)dx$$

2 dimensions

$$\int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx$$

$$g(x) = \int_{y_a}^{y_b} f(x,y)dy \quad \Rightarrow \int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx = \int_{x_a}^{x_b} g(x)dx$$

Quadrature

2 dimensions

$$\int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx$$

$$g(x) = \int_{y_a}^{y_b} f(x,y)dy \quad \Rightarrow \int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx = \int_{x_a}^{x_b} g(x)dx$$

Approximation (4 trapezoidal steps)

$$g(x) \approx \frac{(y_b - y_a)}{4}(f(x,y_a) + 2f(x,3y_a + y_b) + 2f(x,2y_a + 2y_b) + 2f(x,y_a + 3y_b) + f(x,y_b))$$

$$\int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx \approx \frac{(x_b - x_a)}{4}(g(x_a) + 2g(3x_a + x_b) + 2g(2x_a + 2x_b) + 2g(x_a + 3x_b) + g(x_b))$$

Quadrature

2 dimensions

$$\int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx$$

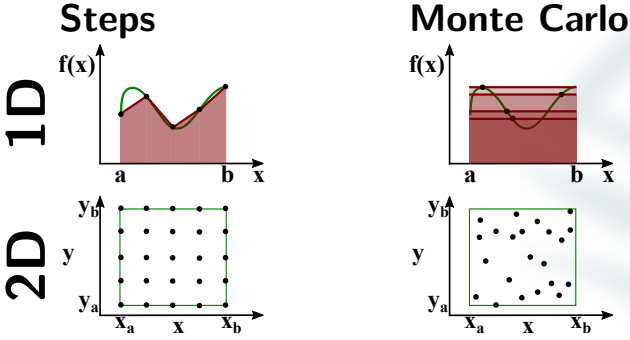
$$g(x) = \int_{y_a}^{y_b} f(x,y)dy \quad \Rightarrow \int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx = \int_{x_a}^{x_b} g(x)dx$$

Approximation (4 Monte Carlo samples)

$$\int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx \approx \frac{(x_b - x_a)(y_b - y_a)}{4}\sum_{i=1}^{4} f(x_i, y_i)$$

$x_i \in [x_a, x_b]$ and $y_i \in [y_a, y_b]$ are uniform random samples.

Quadrature

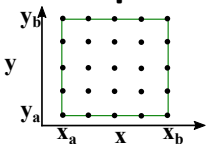


$$\int_a^b f(x)dx$$

$$\int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx$$

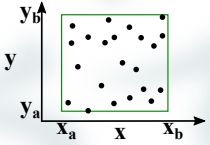
Quadrature

Steps



- Exponential growth
- Curse of dimensionality
- Banding

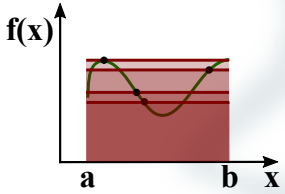
Monte Carlo



- Controlled growth
- Controlled dimensions
- High-frequency noise

Importance sampling

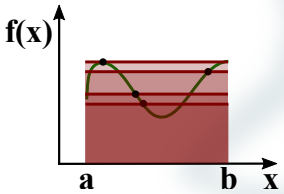
Importance sampling



$$\int_a^b f(x)dx \approx \frac{b-a}{N}\sum_{i=1}^{N} f(x_i)$$

- $N \rightarrow$ number of samples.
- $x_i \in [a, b] \rightarrow$ uniform random samples.

Importance sampling



$$\int_a^b f(x)dx \approx \frac{1}{N}\sum_{i=1}^{N}\frac{f(x_i)}{p(x_i)}$$

- $N \rightarrow$ number of samples.
- $x_i \in [a, b] \rightarrow$ are random samples chosen according to $p(x)$
- $p(x)$ is the *probability distribution function*.

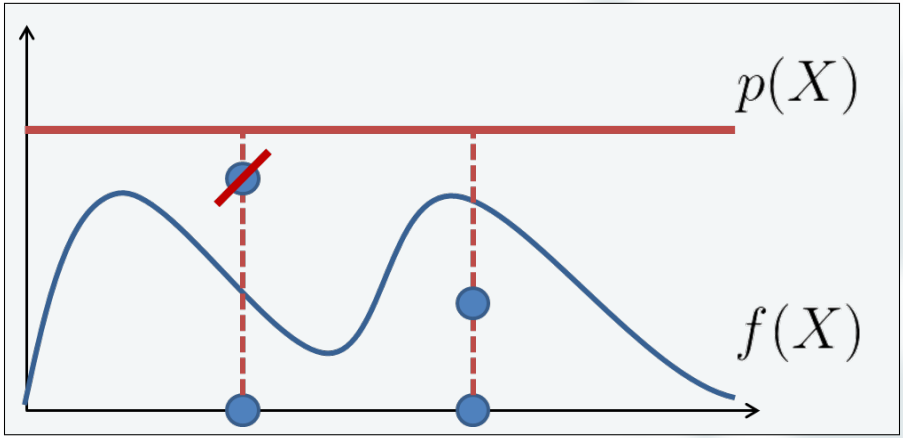
Importance sampling

$$\int_a^b f(x)dx \approx \frac{1}{N}\sum_{i=1}^{N}\frac{f(x_i)}{p(x_i)}$$

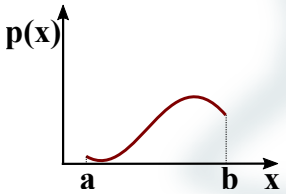
$x_i \in [a, b] \rightarrow$ are random samples chosen according to $p(x)$. How do we do that?

- Inversion method
- Rejection sampling

Rejection sampling



Inversion method



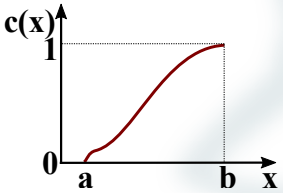
Probability density function (pdf)

$$\int_a^b p(x)dx = 1$$

$$p(x) \geq 0 \quad \forall x \in [a, b]$$

$$p(x) > 0 \quad \text{if} \quad f(x) > 0$$

Inversion method



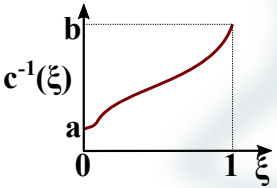
Cumulative density function (cdf)

$$c(x) = \int_a^x p(t)dt$$

$$c(a) = 0$$

$$c(b) = 1$$

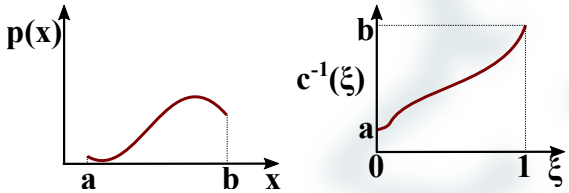
Inversion method



Inverse cdf

$$c^{-1}(0) = a$$
$$c^{-1}(1) = b$$

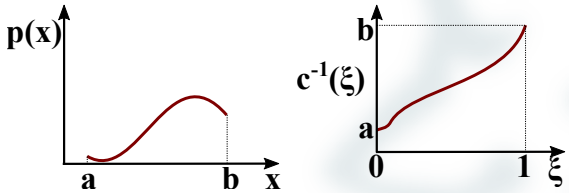
Inversion method



Algorithm

1. Choose N uniform random numbers $\xi_i \in [0, 1)$.
2. Calculate all random samples $x_i = c^{-1}(\xi_i)$.
3. Approximate the integral $\int_a^b f(x)dx \approx \frac{1}{N} \sum_{i=1}^{N} \frac{f(x_i)}{p(x_i)}$.

Inversion method

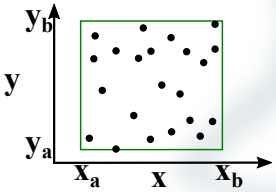


It can be also seen as:

$$\int_a^b f(x)dx \approx \frac{1}{N}\sum_{i=1}^{N}\frac{f\left(c^{-1}\left(\xi_i\right)\right)}{p\left(c^{-1}\left(\xi_i\right)\right)}$$

where $\xi_i \in [0,1) \quad \forall i \in [1,N]$ are uniform random numbers.

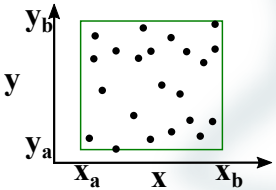
In two dimensions



$$\int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx \approx \frac{1}{N}\sum_{i=1}^{N}\frac{f(x_i,y_i)}{p(x_i,y_i)}$$

- $N \rightarrow$ number of samples.
- $x_i \in [x_a, x_b], y_i \in [y_a, y_b] \rightarrow$ random samples according to $p(x,y)$
- $p(x,y)$ is the 2D *probability distribution function*.

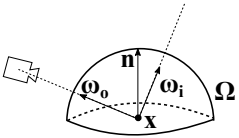
In two dimensions

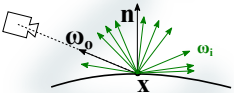


$$\int_{x_a}^{x_b}\int_{y_a}^{y_b} f(x,y)dydx \approx \frac{1}{N}\sum_{i=1}^{N}\frac{f(x_i,y_i)}{p(x_i)p(y_i)}$$

- $N \rightarrow$ number of samples.
- $x_i \in [x_a, x_b], y_i \in [y_a, y_b] \rightarrow$ random samples according to $p(x)$ and $p(y)$
- $p(x)$ and $p(y)$ are two independent *probability distribution functions*.

Sampling strategies





$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) \cos\theta_i \sin\theta_i d\theta_i d\phi_i$$

Sampling strategies

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) \cos\theta_i \sin\theta_i d\theta_i d\phi_i$$

Uniform angle sampling

$$p(\theta_i) = \frac{2}{\pi} \qquad p(\phi_i) = \frac{1}{2\pi}$$

$$c(\theta_i) = \frac{2}{\pi}\theta_i \qquad c(\phi_i) = \frac{1}{2\pi}\phi_i$$

$$c^{-1}(\xi_{\theta_i}) = \frac{\pi}{2}\xi_{\theta_i} \qquad c^{-1}(\xi_{\phi_i}) = 2\pi\xi_{\phi_i}$$

Sampling strategies

$$L_o(\mathbf{x}, \omega_\mathbf{o}) \approx \frac{1}{N} \sum_{i=1}^{N} \frac{2\pi^2 L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) \cos\theta_i \sin\theta_i}{2}$$

Uniform angle sampling

$$c^{-1}(\xi_{\theta_i}) = \frac{\pi}{2}\xi_{\theta_i} \qquad c^{-1}(\xi_{\phi_i}) = 2\pi\xi_{\phi_i}$$

Sampling strategies

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) \cos\theta_i \sin\theta_i d\theta_i d\phi_i$$

Uniform solid angle sampling

$$p(\theta_i) = \sin\theta_i$$
$$c(\theta_i) = \cos\theta_i$$
$$c^{-1}(\xi_{\theta_i}) = \arccos\xi_{\theta_i}$$

$$p(\phi_i) = \frac{1}{2\pi}$$
$$c(\phi_i) = \frac{1}{2\pi}\phi_i$$
$$c^{-1}(\xi_{\phi_i}) = 2\pi\xi_{\phi_i}$$

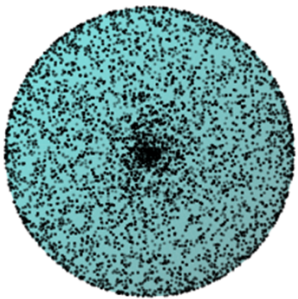
Sampling strategies

$$L_o(\mathbf{x}, \omega_\mathbf{o}) \approx \sum_{i=1}^{N} \frac{2\pi L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) \cos\theta_i \cancel{\sin\theta_i}}{\cancel{\sin\theta_i}}$$

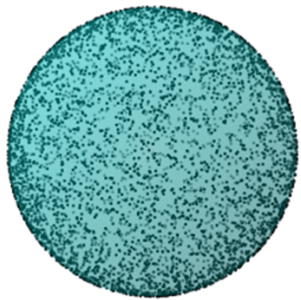
Uniform solid angle sampling

$$c^{-1}(\xi_{\theta_i}) = \arccos \xi_{\theta_i} \qquad c^{-1}(\xi_{\phi_i}) = 2\pi\xi_{\phi_i}$$

Incorrectly distributed - Top View



Correctly distributed - Top View



Sampling strategies

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) \cos\theta_i \sin\theta_i d\theta_i d\phi_i$$

Uniform cosine sampling

$$p(\theta_i) = 2 \sin\theta_i \cos\theta_i$$
$$c(\theta_i) = \sin^2\theta_i$$
$$c^{-1}(\xi_{\theta_i}) = \arcsin\sqrt{\xi_{\theta_i}}$$

$$p(\phi_i) = \frac{1}{2\pi}$$
$$c(\phi_i) = \frac{1}{2\pi}\phi_i$$
$$c^{-1}(\xi_{\phi_i}) = 2\pi\xi_{\phi_i}$$

Sampling strategies

$$L_o(\mathbf{x}, \omega_\mathbf{o}) \approx \sum_{i=1}^{N} \frac{2\pi L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) \cos\theta_i \sin\theta_i}{2 \sin\theta_i \cos\theta_i}$$

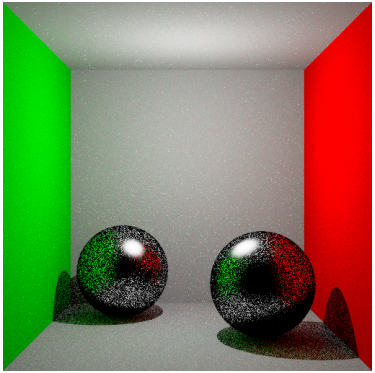
Uniform cosine sampling

$$c^{-1}(\xi_{\theta_i}) = \arcsin\sqrt{\xi_{\theta_i}}$$

$$c^{-1}(\xi_{\phi_i}) = 2\pi\xi_{\phi_i}$$

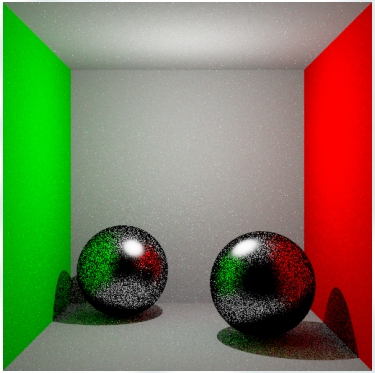
Sampling strategies

Uniform solid angle



8 spp

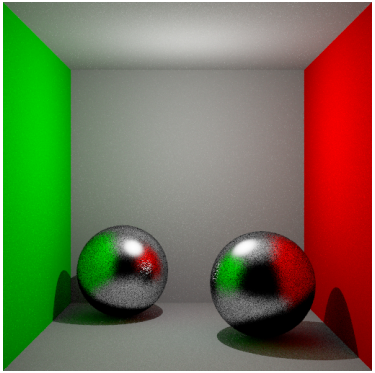
Uniform cosine



8 spp

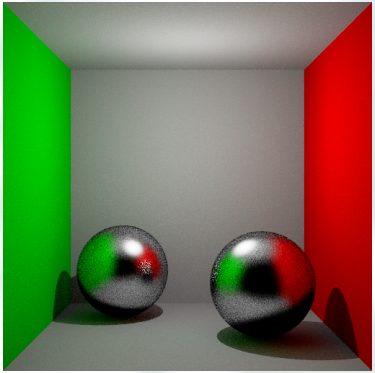
Sampling strategies

Uniform solid angle



128 spp

Uniform cosine

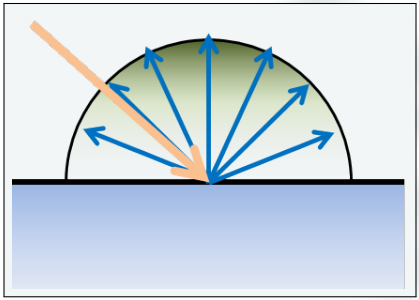


128 spp

Material lobes

About materials

- Diffuse materials ($f_r = k_d/\pi$) are better sampled with cosine sampling.



About materials

- $\mathbf{T}$ is the local to global change of base.
- Generate two random numbers $\xi_{\theta_i} \in [0,1]$ and $\xi_{\phi_i} \in [0,1]$.

$$\theta_i = \arcsin \sqrt{\xi_{\theta_i}}$$
$$\phi_i = 2\pi \xi_{\phi_i}$$
$$\omega_{\mathbf{i}} = \mathbf{T} \begin{pmatrix} \sin\theta_i \cos\phi_i \\ \sin\theta_i \sin\phi_i \\ \cos\theta_i \end{pmatrix}$$
$$L_o(\mathbf{x}, \omega_{\mathbf{o}}) \approx k_d L_i(\mathbf{x}, \omega_{\mathbf{i}})$$


Diffuse

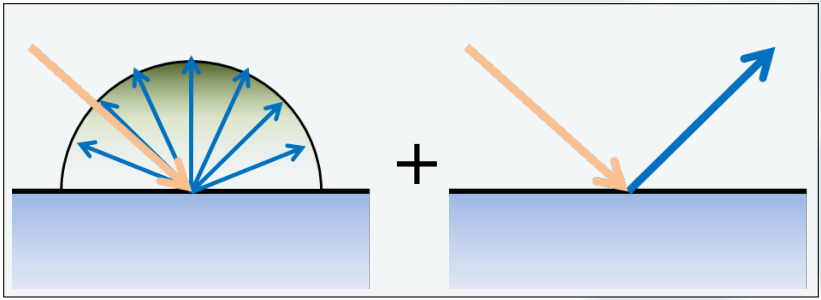
About materials

- Diffuse materials are better sampled with cosine sampling.
- Delta BSDFs (reflection and refraction) are sampled deterministically.

Plastic

Example: plastic as diffuse BRDF plus a delta specular reflection

$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = k_d \frac{1}{\pi} + k_s \frac{\delta_{\omega_\mathbf{r}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}} \qquad \omega_\mathbf{r} = 2\mathbf{n}(\mathbf{n} \cdot \omega_\mathbf{i}) - \omega_\mathbf{i}$$

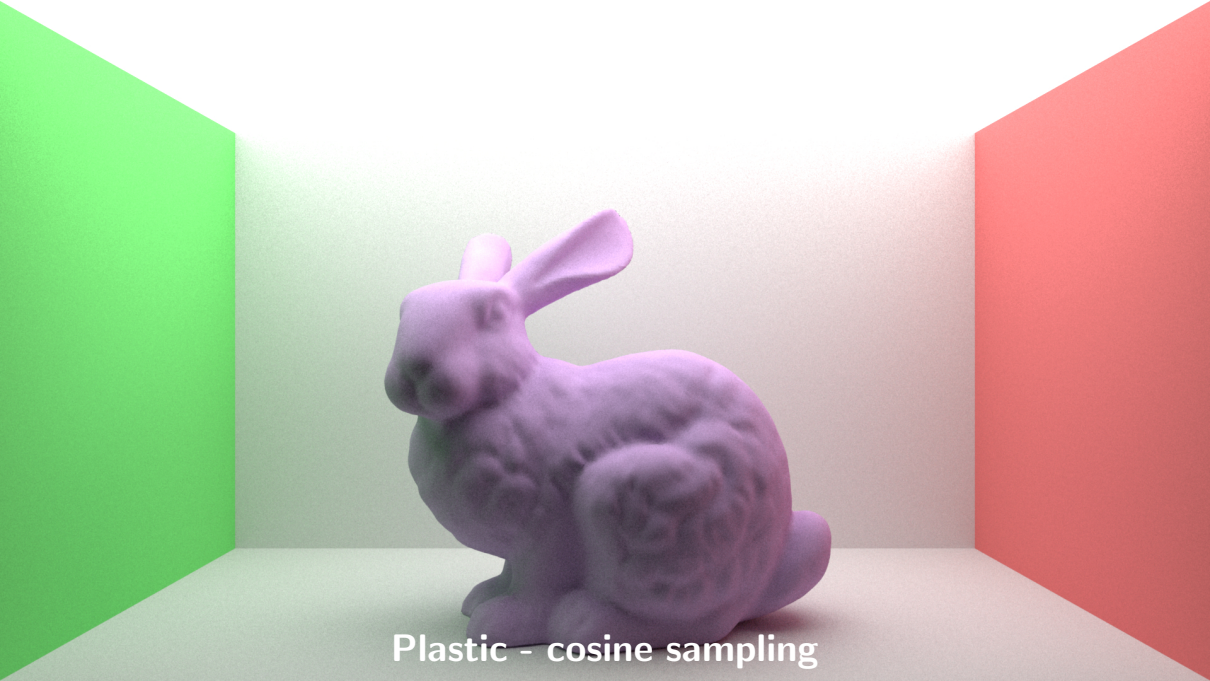


Plastic

Cosine sampling for plastic (or any BSDF):

- $\mathbf{T}$ is the local to global change of base.
- Generate two random numbers $\xi_{\theta_i} \in [0,1]$ and $\xi_{\phi_i} \in [0,1]$.

$$\theta_i = \arccos \sqrt{1 - \xi_{\theta_i}}$$
$$\phi_i = 2\pi \xi_{\phi_i}$$
$$\omega_{\mathbf{i}} = \mathbf{T} \begin{pmatrix} \sin\theta_i \cos\phi_i \\ \sin\theta_i \sin\phi_i \\ \cos\theta_i \end{pmatrix}$$
$$L_o(\mathbf{x}, \omega_{\mathbf{o}}) \approx \pi L_i(\mathbf{x}, \omega_{\mathbf{i}}) f_r(\mathbf{x}, \omega_{\mathbf{i}}, \omega_{\mathbf{o}})$$


Plastic - cosine sampling

Plastic

What happened? Where is the plastic bunny?

Plastic

What happened? Where is the plastic bunny?

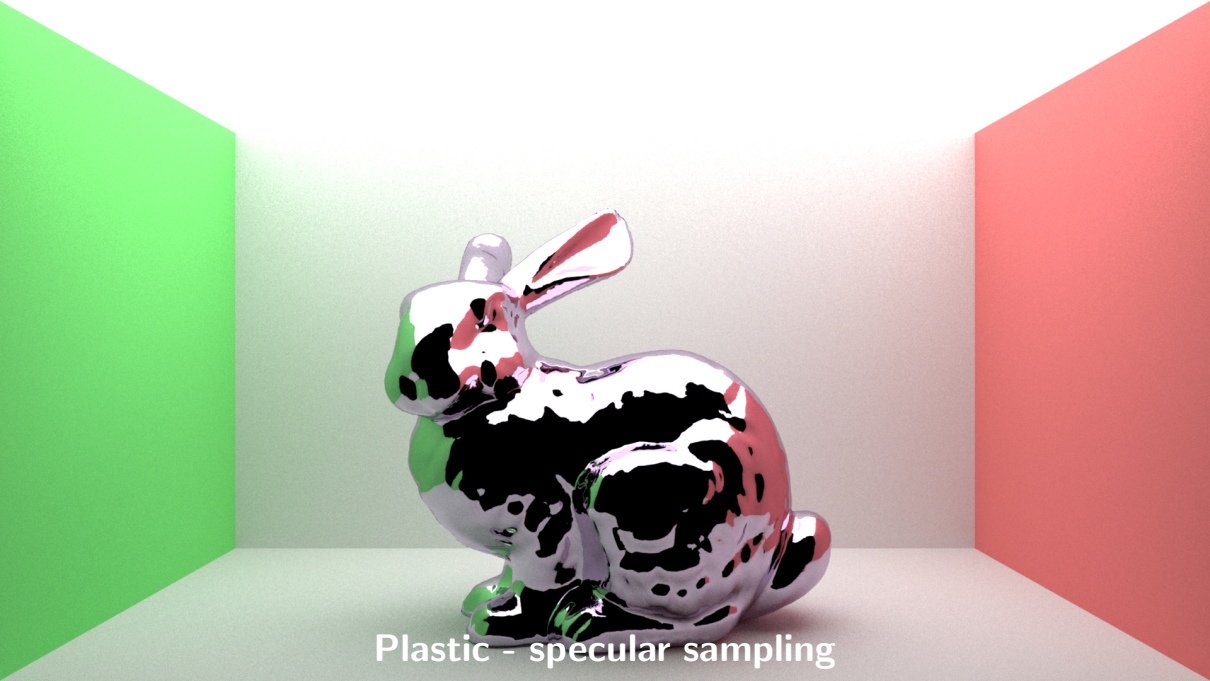
- The chances of sampling exactly the specular direction $\omega_{\mathbf{r}}$ are 0.
- The specular delta part is never found.
- Other specular lobes have also low chances of being found.

Plastic

What happened? Where is the plastic bunny?

- The chances of sampling exactly the specular direction $\omega_{\mathbf{r}}$ are 0.
- The specular delta part is never found.
- Other specular lobes have also low chances of being found.

Well, lets sample the specular direction instead ($\omega_{\mathbf{i}} = \omega_{\mathbf{r}}$).


Plastic - specular sampling

Plastic

What happened now?

Plastic

What happened now?

- No sampled directions account for the diffuse part of the BSDF.
- The diffuse part of the BSDF is ignored.

Plastic

Lets sample both and add both together!

Plastic

Lets sample both and add both together!

- Could do, but we are not acocunting for the relative importance between lobes.
- Sample growth depending on the number of lobes.
- Consider lobes as different samplable events.

Samplable lobes

$$\int_R f_1(x) + f_2(x) + ... + f_n(x)dx = \int_R \sum_{i=1}^{n} f_i(x)dx$$

Each $f_i(x)$ has its own sampling strategy ($p_i(x)$ and $c_i^{-1}(\xi)$).

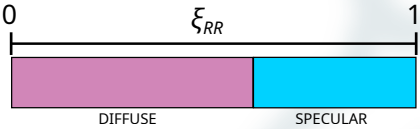
$$\int_R \sum_{i=1}^{n} f_i(x)dx \approx \frac{1}{N} \sum_{j=1}^{N} e_{RR}\left(\xi_{2j-1}, \xi_{2j}\right)$$

where $\xi_i \in [0, 1) \quad \forall i \in [1, 2N]$ are uniform random numbers.

Samplable lobes

$$eRR(\xi_{RR}, \xi_s) = \begin{cases} \dfrac{f_1\left(c_1^{-1}(\xi_s)\right)}{k_1 p_1\left(c_1^{-1}(\xi_s)\right)} & \text{if } \xi_{RR} \in [0, k_1) \\ \dots & \\ \dfrac{f_i\left(c_i^{-1}(\xi_s)\right)}{k_i p_i\left(c_i^{-1}(\xi_s)\right)} & \text{if } \xi_{RR} \left[\sum_{j=1}^{i-1} k_j, \sum_{j=1}^{i} k_j\right) \\ \dots & \\ 0 & \text{if } \xi_{RR} \in \left[\sum_{j=1}^{n} k_j, 1\right) \end{cases}$$

Samplable lobes

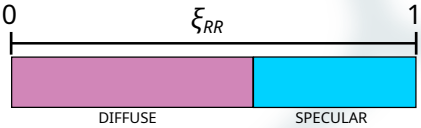


Assign probabilities to each event:

- $p_d = \frac{\max(k_d)}{\max(k_d)+\max(k_s)}$ for the diffuse event.
- $p_s = 1 - p_d$ for the specular event.

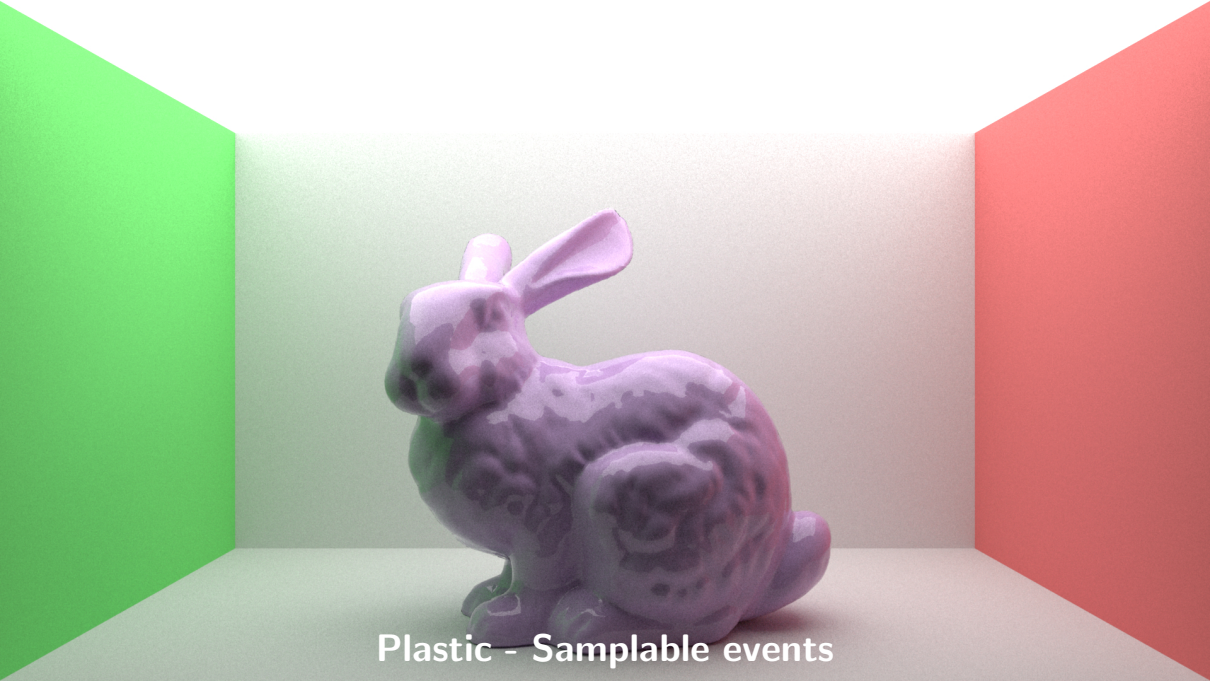
Generate a random number $\xi_{RR} \in [0, 1]$ to choose between both events.

Samplable lobes



Generate random number $\xi_{RR} \in [0, 1]$.

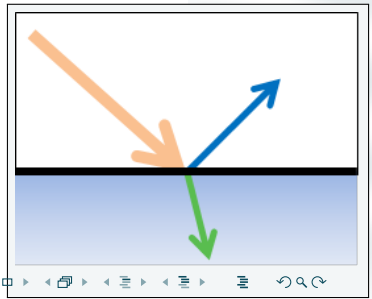
- If $\xi_{RR} \leq p_d$
  - Generate $\omega_\mathbf{i}$ with uniform cosine sampling.
  - $L_o(\mathbf{x}, \omega_\mathbf{o}) \approx L_i(\mathbf{x}, \omega_\mathbf{i}) \frac{k_d}{p_d}$
- If $\xi_{RR} > p_d$
  - Generate $\omega_\mathbf{i} = \omega_\mathbf{r}$.
  - $L_o(\mathbf{x}, \omega_\mathbf{o}) \approx L_i(\mathbf{x}, \omega_\mathbf{i}) \frac{k_s}{p_s}$


Plastic – Samplable events

Dielectric

Dielectrics: delta reflection ($\omega_\mathbf{r}$ from **Law of Reflection**) plus delta refraction ($\omega_\mathbf{t}$ from to **Sahl-Snell's Law**).

$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = k_s(\omega_\mathbf{i}) \frac{\delta_{\omega_\mathbf{r}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}} + k_t(\omega_\mathbf{i}) \frac{\delta_{\omega_\mathbf{t}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}}$$



Dielectric

$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = k_s(\omega_\mathbf{o}) \frac{\delta_{\omega_\mathbf{r}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}} + k_t(\omega_\mathbf{o}) \frac{\delta_{\omega_\mathbf{t}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}}$$

$k_s(\omega_\mathbf{o})$ and $k_t(\omega_\mathbf{o})$ are not constant, the depend on the **Fresnel Equations**

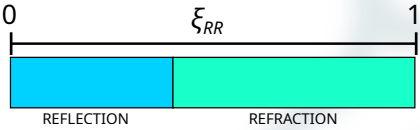
$$\rho_{\parallel} = \frac{\eta_1 \cos\theta_i - \eta_0 \cos\theta_t}{\eta_1 \cos\theta_i + \eta_0 \cos\theta_t}$$

$$\rho_{\perp} = \frac{\eta_0 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_0 \cos\theta_i + \eta_1 \cos\theta_t}$$

reflected: $F_r = \frac{1}{2}\left(\rho_{\parallel}^2 + \rho_{\perp}^2\right)$

refracted: $F_t = 1 - F_r$

Dielectric

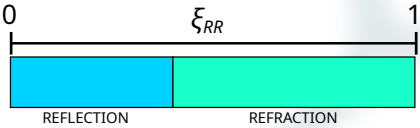


Assign probabilities to each event:

- $p_s = \frac{\max(k_s)}{\max(k_s)+\max(k_t)}$ for the reflection event.
- $p_t = 1 - p_s$ for the specular event.

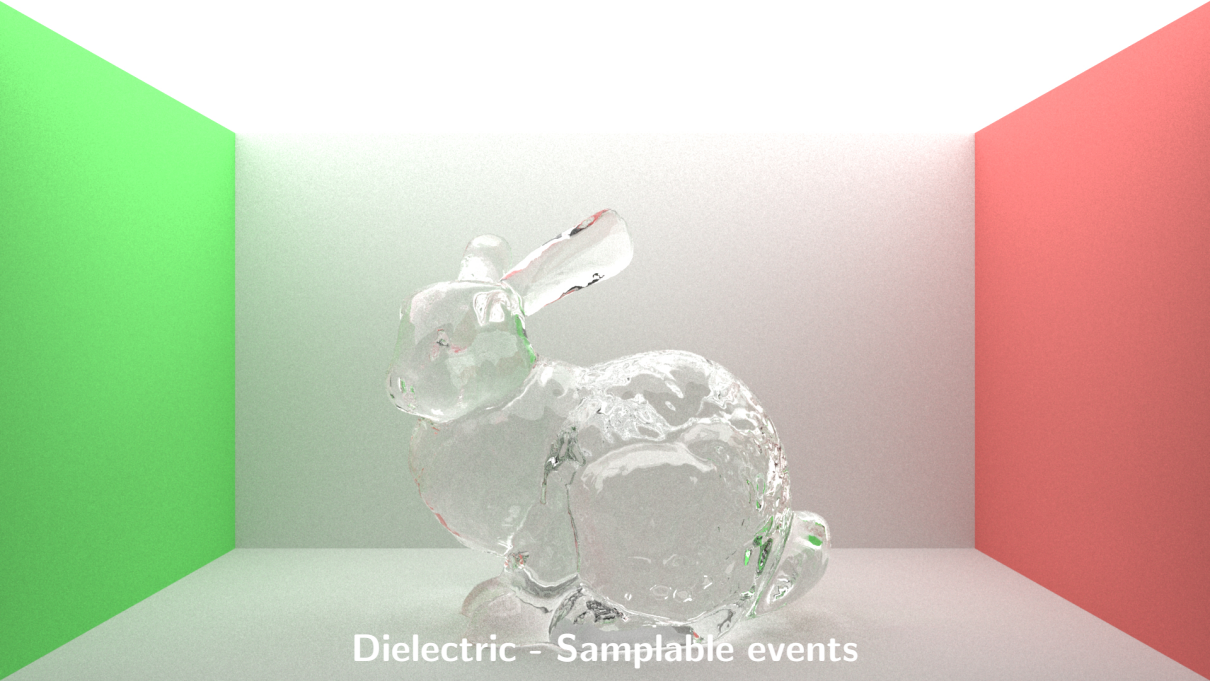
Generate a random number $\xi_{RR} \in [0, 1]$ to choose between both events.

Dielectric



Generate random number $\xi_{RR} \in [0, 1]$.

- If $\xi_{RR} \leq p_s$
  - Generate $\omega_\mathbf{i} = \omega_\mathbf{r}$ with uniform cosine sampling.
  - $L_o(\mathbf{x}, \omega_\mathbf{o}) \approx L_i(\mathbf{x}, \omega_\mathbf{i}) \frac{k_s}{p_s}$
- If $\xi_{RR} > p_s$
  - Generate $\omega_\mathbf{i} = \omega_\mathbf{t}$.
  - $L_o(\mathbf{x}, \omega_\mathbf{o}) \approx L_i(\mathbf{x}, \omega_\mathbf{i}) \frac{k_t}{p_t}$


Dielectric - Samplable events

Material model

Do you have to choose the sampling strategy according to the material's type?

Material model

Do you have to choose the sampling strategy according to the material's type?

No! You can combine all material lobes into a single BSDF representation.

$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = k_d \frac{1}{\pi} + k_s \frac{\delta_{\omega_\mathbf{r}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}} + k_t \frac{\delta_{\omega_\mathbf{t}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}}$$

and then the probability of the event can become zero ($\max(k_l) = 0$ means $p_l = 0$).

Material model

In the general case:

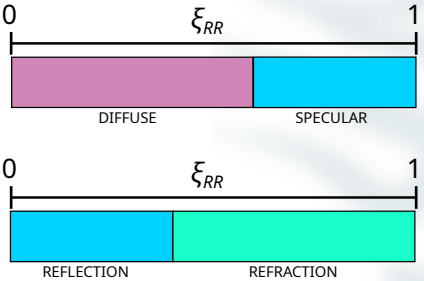
$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = \sum_l k_l(\mathbf{x}, \omega_\mathbf{o}) f_l(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o})$$

We include the interaction point $\mathbf{x}$ and the outgoing direction $\omega_\mathbf{o}$ in the coefficients in order to account for both textures and Fresnel-like effects.

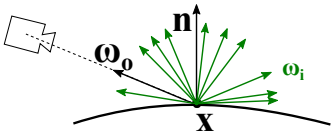
Material model

$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = k_d \frac{1}{\pi} + k_s(\mathbf{x}, \omega_\mathbf{o}) \frac{\delta_{\omega_\mathbf{r}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}} + k_t(\mathbf{x}, \omega_\mathbf{o}) \frac{\delta_{\omega_\mathbf{t}}(\omega_\mathbf{i})}{\mathbf{n} \cdot \omega_\mathbf{i}}$$

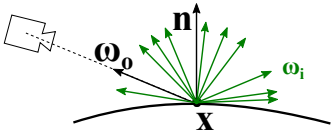
Zero probability discards unused lobes ($\max(k_l) = 0 \Rightarrow p_l = 0$).

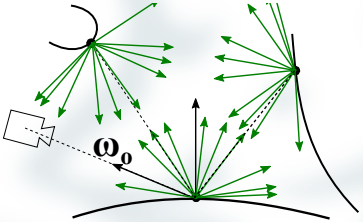


Stopping criterion



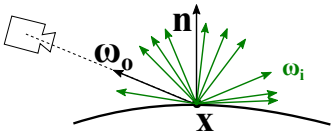
Stopping criterion

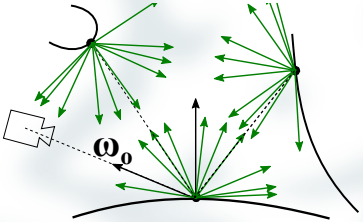




When do we stop?

Stopping criterion





When do we stop?

- Limit the number of bounces.
- Absorption event: assign a probability to not doing anything (Russian Roulette).

Monte Carlo

Informática Gráfica



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Grado en Ingeniería Informática

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